

SALEM AHMED

AI/ML Enthusiast · Researcher · Instructor · Lead Organizer

+20 114 271 2328 | salemsalemahmed761@gmail.com | [LinkedIn](#) | [GitHub](#)

PROFESSIONAL SUMMARY

High-achieving AI/ML engineer and STEM educator with national-level competitive accolades, hands-on research experience in intelligent tutoring systems, and a proven record of technical leadership. Gold Medalist at the Egyptian Olympiad in Computer Science (EOCS) and National Finalist at ISEF with a plant-disease detection AI project. Passionate about building scalable, real-world AI solutions and mentoring the next generation of scientists.

COMPETITIVE ACHIEVEMENTS

Gold Medal — Egyptian Olympiad in Computer Science (EOCS)

National algorithmic programming competition; solutions implemented in Python and C++.

National Finalist — Intel ISEF (International Science & Engineering Fair)

Developed and presented a real-time plant disease detection system leveraging deep learning; selected among top national research projects.

TECHNICAL SKILLS

Languages	Python, C++, JavaScript, Dart
AI / ML	Machine Learning, Deep Learning, Neural Networks, Prompt Engineering, Neuro-Symbolic AI, RAG (Retrieval-Augmented Generation), Bayesian Inference
Backend	Backend Development, REST APIs, Data Analysis & Pipelines
Tools & IDEs	Git/GitHub, MS Office 365, Microsoft Excel, PowerPoint
Methodologies	ATS-Optimized Writing, Scientific Writing, AI Ethics & Model Evaluation, Agile/Project Management
Soft Skills	Technical Leadership, Mentorship, Critical Thinking, Public Speaking, Research, Teamwork

FEATURED RESEARCH PROJECT

The CALM System — *Cognitive Apprenticeship via Neuro-Symbolic AI*

- ▶ Architected an AI tutoring system fusing LLM fluency with neuro-symbolic verification to deliver PhD-level STEM mentorship — targeting 300M+ underserved learners at zero marginal cost.
- ▶ Engineered a tripartite inference pipeline: Retrieval-Augmented Generation (RAG) for factual grounding, Socratic scaffolding for pedagogy, and Bayesian mastery tracking for adaptive learning.
- ▶ Motivated by Bloom's 2-Sigma Problem; designed to achieve ≥ 2.0 SD uplift in student performance. Validation protocol: randomized controlled trial (N = 120) targeting effect size $d = 1.2$ – 1.8 across physics, mathematics, and reasoning domains.
- ▶ Plant Disease Detection (ISEF): Built a real-time deep learning inference pipeline for instantaneous plant pathology diagnosis; reached national finals in Egypt's most prestigious science competition.

LEADERSHIP & PROFESSIONAL EXPERIENCE

Lead Organizer & Program Coordinator · [TelqAI Program](#) *Dec 2025 – Present*

- ▶ Spearhead organizational operations for an AI-focused program, coordinating cross-functional teams and managing program delivery at scale.
- ▶ Design and implement structured workflows to ensure program milestones are met on schedule.

Instructor · [Bootcamp](#) *Sep 2025 – Dec 2025*

- ▶ Delivered intensive technical instruction to cohorts of students, covering AI/ML fundamentals, programming, and problem-solving methodologies.
- ▶ Developed original curriculum and hands-on project modules, increasing student engagement and knowledge retention.

Junior Researcher · [Youth Science Journal](#) *Aug 2025 – Dec 2025*

- ▶ Authored and peer-reviewed research articles on emerging AI and computer science topics; maintained scientific rigor and editorial standards.
- ▶ Collaborated with senior researchers to synthesize complex technical findings into accessible, publication-ready content.

Head of ROS (Robotics Operating System) · [Robotics Club](#) *Ongoing*

- ▶ Lead the ROS division, mentoring club members in robotics software integration, sensor fusion, and autonomous system design.
- ▶ Organised workshops and hands-on challenges to advance members' technical proficiency in embedded systems and AI robotics.

Co-Head of Machine Learning · [Model Arab League \(MAL\)](#) *Dec 2025 – Present*

- ▶ Co-direct the ML division within a pan-Arab simulation and leadership organization; guide members on AI policy, ethics, and technical implementation.

High Board Management · [CS Club](#) *Dec 2025 – Present*

- ▶ Serve on the executive board overseeing club strategy, event programming, and member development for a competitive computer science community.

EDUCATION

STEM High School For Boys *In Progress*

Diploma — Science, Technology, Engineering & Mathematics

LANGUAGES

Arabic (Native) · **English** (Professional) · **Deutsch**(Conversational)